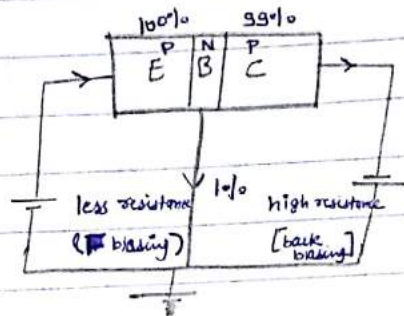


Transistor :- Transfer of resistor



Current goes from low resistance to high resistance without diminishing
[∴ Transfer of resistor]

Small signal low frequency Analysis :-

Parameter of defining signals — voltage Amplitude
current —
frequency

→ h parameter is the best suitable for transistor modelling.

2 port
4 terminal
needed
∴ terminals
are made
common

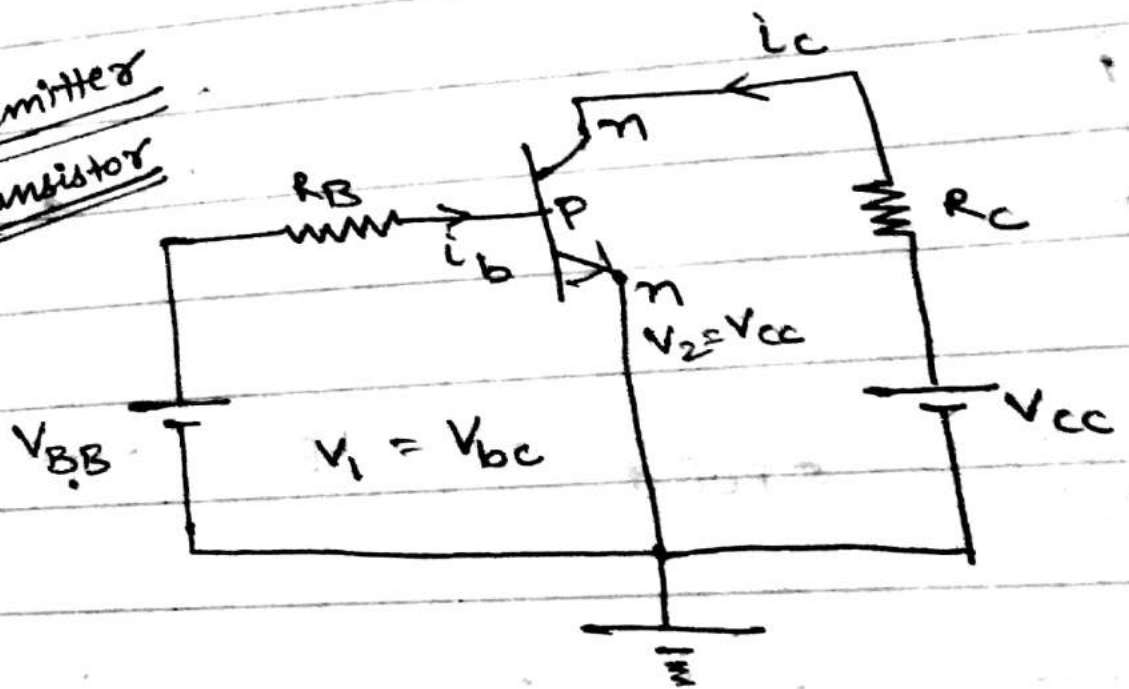
Common emitter — i/p :- base } gives voltage & current
o/p :- collector } gain simultaneously

Common base — i/p :- emitter } does not give
o/p :- collector } current gain

Common collector — i/p :- base } current gain
[buffer] o/p :- emitter

To maintain a voltage level in a digital circuit buffer is used
↓
Common collector
To increase fan out

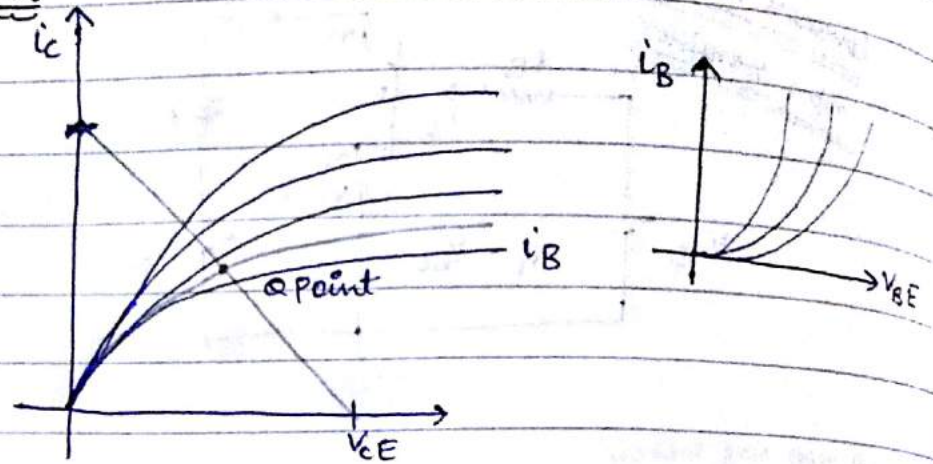
Common emitter
npn transistor



$$V_{be} = h_{11} i_b + h_{12} V_{ce}$$

$$i_c = h_{21} i_b + h_{22} V_{ce}$$

Output characteristic:



V_{BE} = Total Instantaneous value

v_{be} = ac value

V_{be} = rms value of ac value

V_{BE} = DC value / Avg value

$$V_{BE} = v_{be} + V_{BE}$$

$h_{11} \rightarrow$ out put short circuit, i/p resistance of transistor

$$\frac{v_{be}}{i_b} = h_{11} \Big|_{v_{ce}=0}$$

$h_{11} = h_{ie} \rightarrow$ input resistance, common emitter

$$h_{12} = \frac{v_{be}}{v_{ce}} \Big|_{i_b=0}$$

input open circuit

$h_{12} \rightarrow$ reverse voltage gain

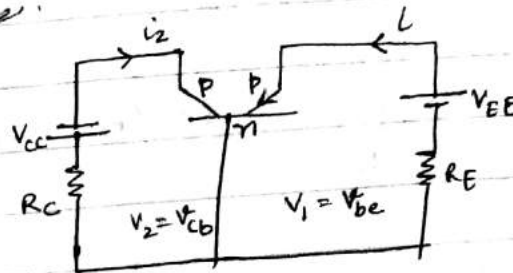
$h_{12} \rightarrow h_{re}$

$h_{21} \rightarrow h_{fe} \rightarrow$ forward current gain

$h_{22} \rightarrow h_{oe} \rightarrow$ o/p admittance

→ with high amplitude, this hybrid model fails due to non-linearity behaviour of the circuit.

Common-Base
PNP configuration



$i_{IP} \rightarrow$

$$\begin{aligned} V_1 &= V_{BE} \\ V_2 &= V_{CB} \\ I_1 &= i_e \\ I_2 &= i_c \end{aligned}$$

$$\begin{aligned} V_{be} &= h_{11} i_e + h_{12} V_{cb} \\ i_c &= h_{21} i_e + h_{22} V_{cb} \end{aligned}$$

✓
V_{be} → 2nd port
input to write

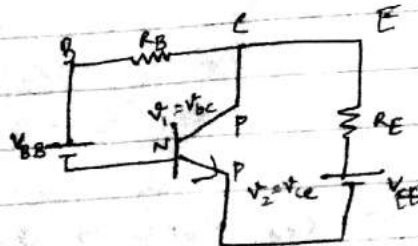
$$h_{11} = h_{ib} = \frac{V_{be}}{i_e} \Big|_{V_{cb}=0} \rightarrow \text{o/p short circuit, i/p resistance}$$

$$h_{12} = h_{rb} = \frac{V_{be}}{V_{cb}} \Big|_{i_e=0} \rightarrow \text{i/p open circuit, reverse voltage gain}$$

$$h_{21} = h_{fb} = \frac{i_c}{i_e} \Big|_{V_{cb}=0} \rightarrow \text{o/p short circuit, forward current gain}$$

$$h_{22} = h_{ob} = \frac{i_c}{V_{cb}} \Big|_{i_e=0} \rightarrow \text{i/p open circuit, o/p admittance}$$

Common-collector
configuration PNP



$$\begin{aligned} I_1 &= i_b \\ I_2 &= i_e \\ V_1 &= V_{BE} \\ V_2 &= V_{CE} \end{aligned}$$

$$\begin{aligned} V_1 &= h_{11} i_1 + h_{12} V_2 \\ I_2 &= h_{21} i_1 + h_{22} V_2 \end{aligned}$$

INPUT
APPROXIMATE
OUTPUT
REVERSE GAIN

$$V_{bc} = h_{11} i_b + h_{12} V_{ce}$$

$$i_c = h_{21} i_b + h_{22} V_{ce}$$

$$h_{11} = \left. \frac{V_{bc}}{i_b} \right|_{V_{ce}=0} \xrightarrow{\text{o/p short circuit i/p resistance}} = h_{ic}$$

$$h_{12} = \left. \frac{V_{bc}}{V_{ce}} \right|_{i_b=0} \xrightarrow{\text{i/p open, circuit voltage gain}}$$

$$h_{21} \rightarrow h_{fc}$$

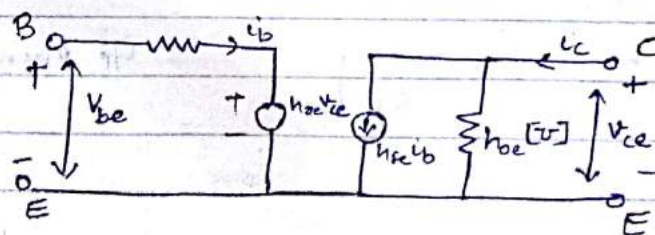
$$h_{22} \rightarrow h_{oc}$$

H-Parameter model :-

Common emitter circuit model -

$$V_{be} = h_{ie} i_b + h_{oe} V_{ce}$$

$$i_c = h_{fe} i_b + h_{oe} V_{ce}$$



→ Models for all the three configurations are inter convertible ←

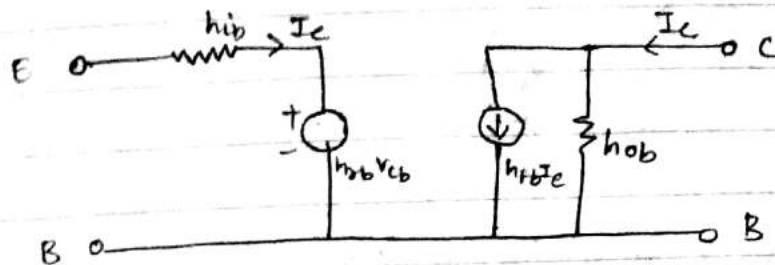
→ To Convert one type of Parameter into Another type :-

Ques - Convert h_{oe} in terms of common base parameter

$$V_{eb} = h_{ib} i_e + h_{rb} V_{cb}$$

$$i_c = h_{fb} i_e + h_{ob} V_{cb}$$

- * Jisko convert karna hai uska formula likho
- * Jisme convert karna hai uska diagram banao



$$h_{oe} = \frac{V_{ce}}{V_{cb}} \Big|_{I_b=0}$$

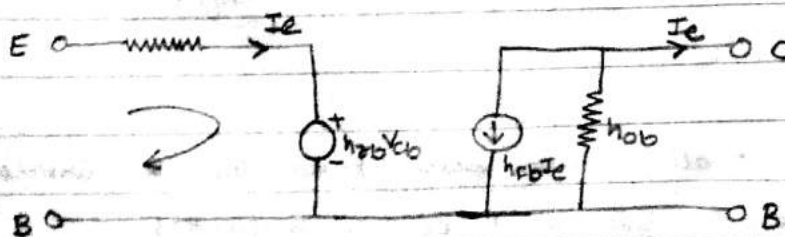
$$h_{oe} = 1 + \frac{V_{bc}}{V_{ce}} \Big|_{I_b=0}$$

- * All the currents are incoming
i.e. varying [R.M.S.]

$$I_e + I_b + I_c = 0$$

$$-I_e = I_c$$

$$\text{or } I_c = -I_e$$



$$E \rightarrow B \rightarrow C \rightarrow E$$

$$V_{eb} + V_{bc} + V_{ce} = 0$$

$$V_{eb} = h_{ib} I_e + h_{fb} V_{cb}$$

$$V_{eb} = h_{ib} I_e - h_{fb} V_{bc}$$

$$h_{ib} I_e - h_{fb} V_{bc} + V_{bc} + V_{ce} = 0$$

$$h_{ib} I_e + (1 - h_{fb}) V_{bc} + V_{ce} = 0$$

$$h_{ib} \frac{I_e}{V_{ce}} + (1 - h_{fb}) \frac{V_{bc}}{V_{ce}} + 1 = 0$$

$$I_e = -h_{fb} I_e - h_{ob} V_{cb}$$

$$= -h_{fb} I_e + h_{ob} V_{bc}$$

$$I_e = \frac{h_{ob}}{1+h_{fb}} V_{bc}$$

$$\frac{h_{ib} h_{ob}}{1+h_{fb}} V_{bc} + (1-h_{ob}) V_{bc} + V_{ce} = 0$$

$$\frac{(h_{ib} h_{ob} + 1 + h_{fb} - h_{ob} - h_{ob} h_{fb})}{1+h_{fb}} V_{bc} + V_{ce} = 0$$

$$\frac{V_{bc}}{V_{ce}} = - \frac{(1+h_{fb})}{(1+h_{fb} + h_{ib} h_{ob} - h_{ob} - h_{ob} h_{fb})}$$

$$h_{re} = \frac{1+h_{fb} + h_{ib} h_{ob} - h_{ob} - h_{ob} h_{fb} - 1 - h_{fb}}{h_{ib} h_{ob} + (1-h_{ob})(1+h_{fb})}$$

$$h_{re} = \frac{h_{ib} h_{ob} - (1+h_{fb}) h_{ob}}{h_{ib} h_{ob} + (1-h_{ob})(1+h_{fb})}$$

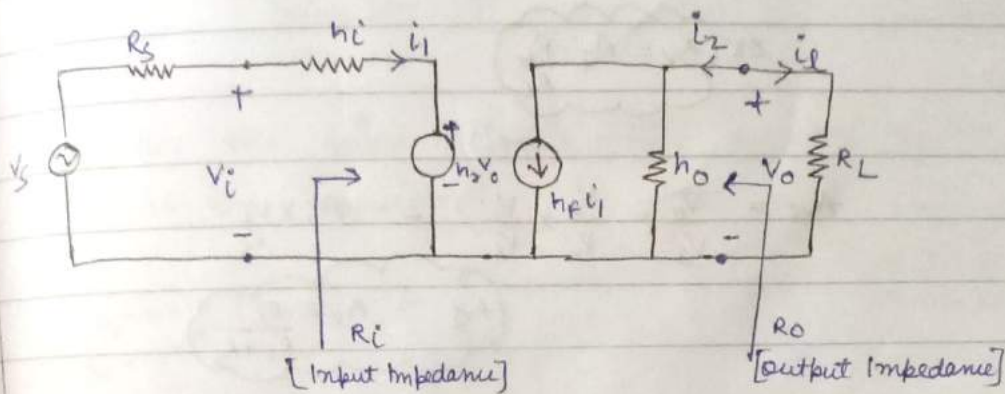
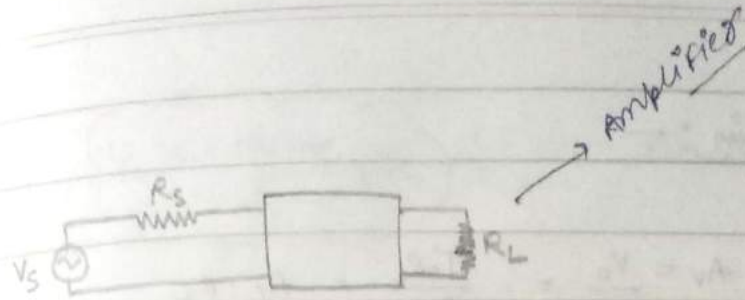
Assignment

Ques - Convert all common base parameter & common collector parameters in terms of CE parameters.

Ques - Convert all the exact hybrid parameter formula into approx. value

Ques - Table 8.3 [learn it]

#



∴ Current gain :-

$$A_i = \frac{i_2}{i_1} = -\frac{i_2}{i_1}$$

$$i_2 = h_f i_1 + h_o V_o$$

$$\text{or } -i_2 = h_f i_1 + h_o V_o$$

$$\frac{-i_2}{i_1} = h_f + h_o \frac{V_o}{i_1} \quad [V_o = i_2 R_L = -i_2 R_L]$$

$$\text{or } -\left(\frac{i_2}{i_1}\right) = h_f + h_o \left(\frac{i_2}{i_1}\right) R_L$$

$$-A_i = h_f + h_o A_i R_L$$

$$A_i = \frac{-h_f}{1 + h_o R_L}$$

∴ Input Impedance :-

$$R_i = \frac{V_i}{i_1} = \frac{h_i i_1 + h_o V_o}{i_1} = \frac{h_i i_1 + h_o i_2 R_L}{i_1}$$

$$R_i = h_i + h_o A_i R_L$$

∴ voltage gain :-

$$A_V = \frac{V_o}{V_i} = \frac{I_e R_L}{I_i R_i} = A_I \cdot \frac{R_L}{R_i}$$

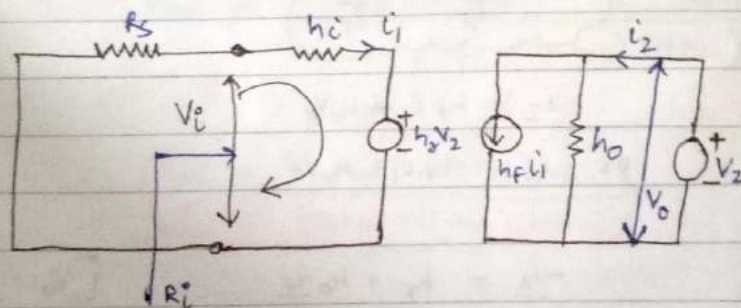
or $A_V = A_I \cdot \frac{R_L}{R_i}$

$$A_{VS} = \frac{V_o}{V_S} = \frac{V_o}{V_i} \times \frac{V_i}{V_S} = A_V \times \frac{V_i}{V_S}$$

$A_{VS} = A_V \times \frac{R_i}{R_i + R_S}$

∴ output impedance :-

$$R_o = \frac{V_o}{I_2} \Big|_{R_2 \rightarrow \infty, V_S = 0}$$



$$Y_o = \frac{1}{R_o} = \frac{I_2}{V_2}$$

$$I_2 = h_f I_1 + h_o V_2$$

$$Y_o = h_f \frac{I_1}{V_2} + h_o$$

$$(R_S + h_i) I_1 + h_r V_2 = 0$$

$$\Rightarrow \frac{I_1}{V_2} = \frac{-h_r}{R_S + h_i}$$

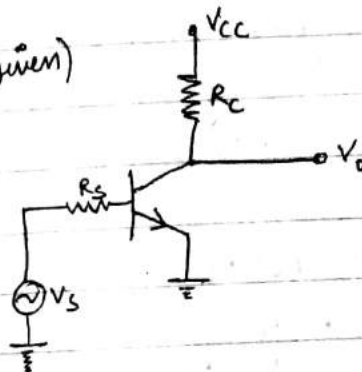
$$Y_o = \frac{-h_f h_r}{R_S + h_i} + h_o$$

$$Y_o = \frac{-h_f h_o}{R_s + h_i} + h_o$$

$$R_o = \frac{1}{Y_o}$$

Ques - Consider the given circuit Find out A_{v_s} , A_v , A_i , R_i , R_o , R_o' of the transistor.

$R_s = 100 \Omega$ (given)
 $R_c = 4 K \Omega$



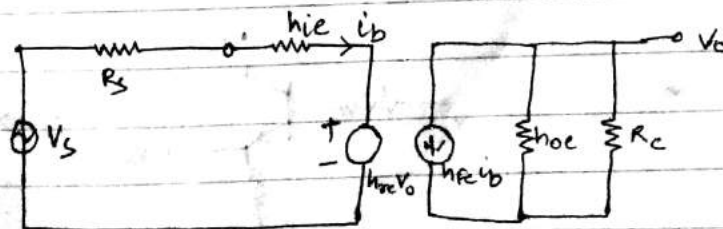
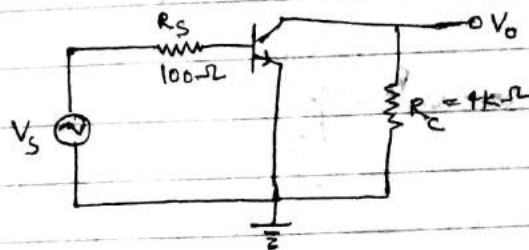
Take the value of h parameters as.

$$h_{ie} = 1.1 K \Omega$$

$$h_{fe} = 50$$

$$h_{re} = 2.5 \times 10^{-4}$$

$$h_{oe} = \frac{1}{40 K \Omega}$$



$$A_i = \frac{-i_c}{i_b} = \frac{-h_{fe}}{1 + h_{oe} R_L} = \frac{-50}{1 + \frac{1}{40 \times 10^3} \times 4 \times 10^3} = -45.45$$

$$R_i = h_{ie} + h_{re} A_i R_L$$

$$= 1.1 \times 10^3 + 2.5 \times 10^{-4} \times (-45.45) \times 4 \times 10^3$$

$$= 1100 - 45.45$$

$$R_i = 1054.55 \text{ or } 1.054 K \Omega$$

$$A_v = A_i \times \frac{R_L}{R_i} = -45.45 \times \frac{4 \times 10^3}{1.08 \times 10^3}$$

$$A_v = -172.49$$

$$A_{vs} = \frac{A_v \cdot R_i}{R_i + R_s}$$

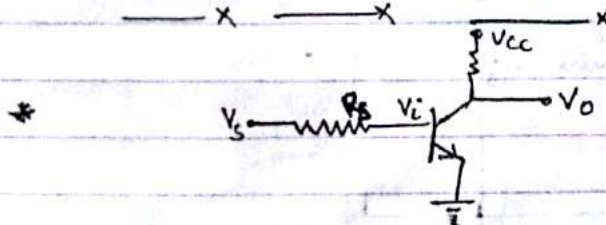
$$A_{vs} = -157.54$$

$$Y_o = \frac{-h_f h_r}{R_s + h_i} + h_o$$

$$R_o = 68.6 \text{ k}\Omega$$

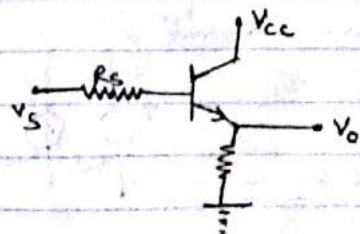
$$R_o' = R_o \parallel R_L$$

$$= \frac{R_o \cdot R_L}{R_o + R_L} = \frac{68.6 \times 4}{72.6} = 3.779 \text{ }\Omega$$



Common emitter

Phase inverse 180° has

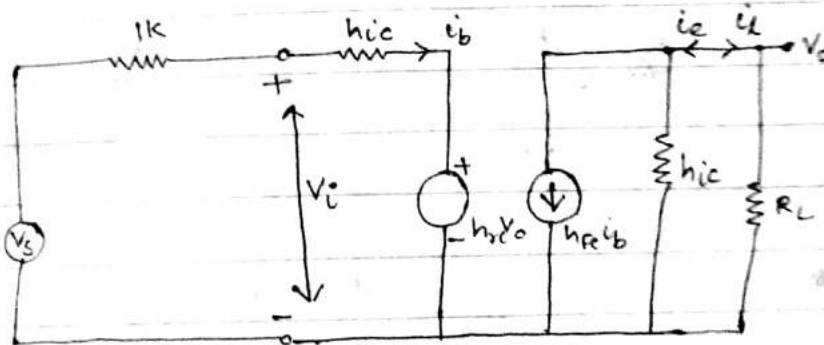
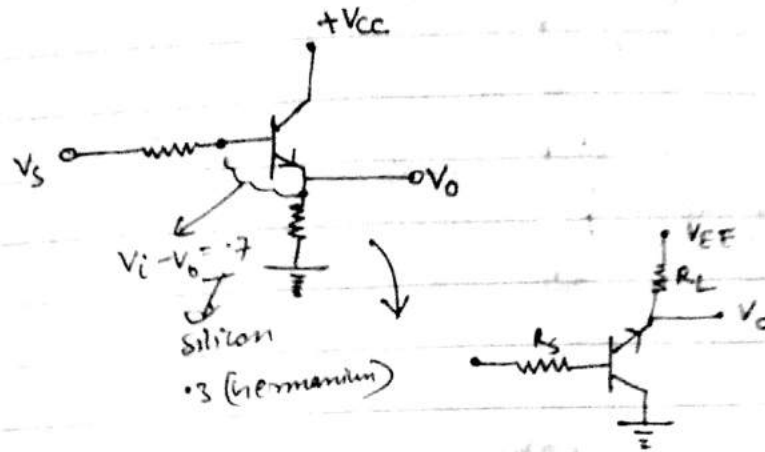


Common collector

Inversion doesn't take

Emmitter follower

Ques-



$$R_L = 4K$$

$$h_{oc} = \frac{1}{40K}$$

$$h_{fc} = -51$$

$$h_{ic} = 1.1K$$

$$h_{rc} = 1$$

$$A_I = +16.36$$

$$R_L = 4K$$

$$A_I = \frac{-i_e}{i_b} = \frac{-h_{fc}}{1+h_{oc}R_L}$$

$$= +16.36$$

$$R_i = h_{ic} + A_I h_{oc} R_L$$

$$R_i = 186.53 K\Omega$$

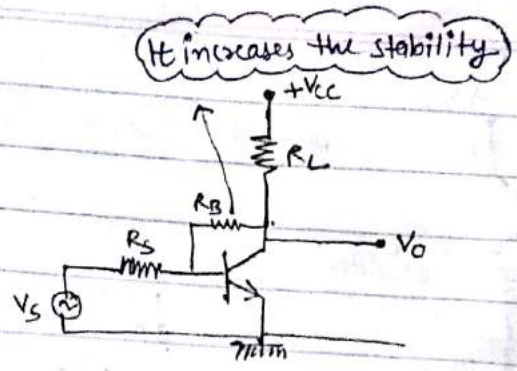
$$A_V = A_I \times \frac{R_L}{R_i} = 0.99$$

$$V_o = h_{oc} - \frac{h_{fc} h_{rc}}{R_s + h_{ic}}$$

$$R_o = \frac{1}{Y_o} = 41.13 \Omega$$

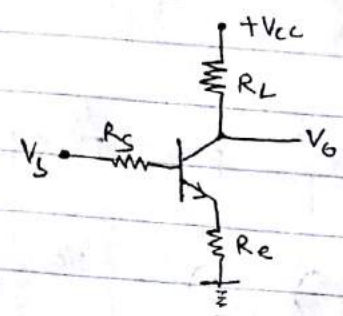
output impedance is very less in common collector

①



To stable the Q-point
 R_B plays a vital role.

②

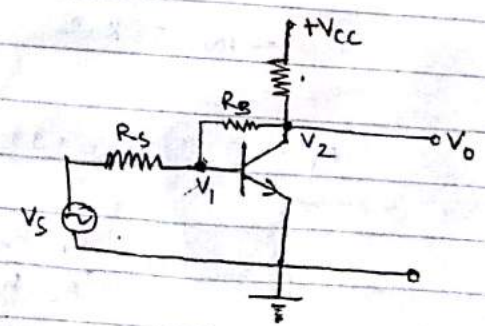


[emitter resistance]
 R_E is included to
stabilise the Q-point.

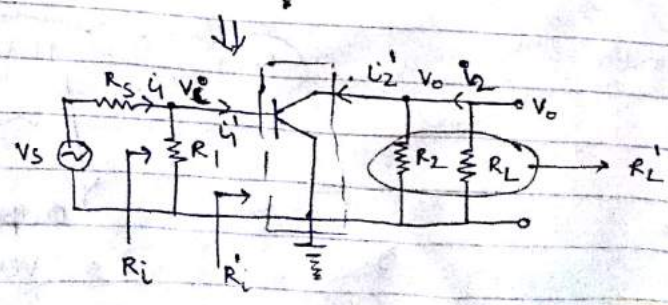
* we can't use direct formula in these 2 circuits *

∴ To overcome the demerits of these circuits: —

- ① Moller's Theorem
- ② Miller's Dwell theorem.



$$\frac{V_2}{V_1} = K = A_v$$



$$R_1 = \frac{R_B}{1-K} \quad R_2 = \frac{R_B}{1-\frac{1}{K}}$$

$$A_I' = -\frac{i_2'}{i_1'} = -\frac{h_{fe}}{1+h_{oe}R_L'}$$

$$R_1' = h_{ie} + A_I' h_{oe} R_L'$$

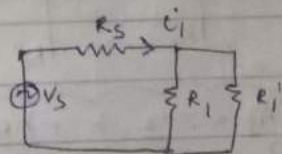
$$A_V = \frac{V_o}{V_i} = \frac{i_2' R_L'}{i_1' R_1'} = \frac{A_I' \cdot R_L'}{R_1'} = K$$

For common emitter;
 $K = A_V \gg 1$

$$R_1 = \frac{R_B}{1-K}$$

$$R_2 = R_B$$

~~$$\frac{i_2}{i_1} = \frac{R_2}{R_1+R_2}$$~~



~~$$\frac{i_1'}{i_1} = \frac{R_1}{R_1+R_1'}$$~~

$$\frac{i_1}{i_2} = \frac{R_1}{(R_1+R_1')} \frac{i_1'}{i_2'}$$

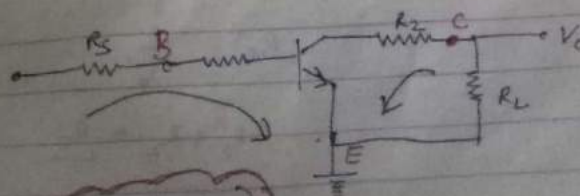
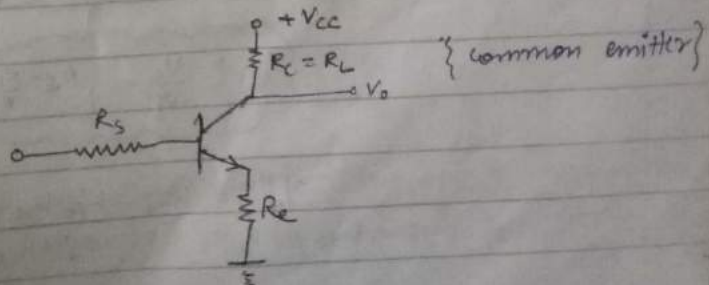
$$A_I = -\frac{i_2}{i_1} = \frac{V_o}{R_L} \times \frac{R_L+R_S}{V_S}$$

$$A_I = -\frac{i_2}{i_1} \times \frac{i_1'}{i_1} \times \frac{i_1'}{i_2'}$$

$$R_i = R_1' \parallel R_1$$

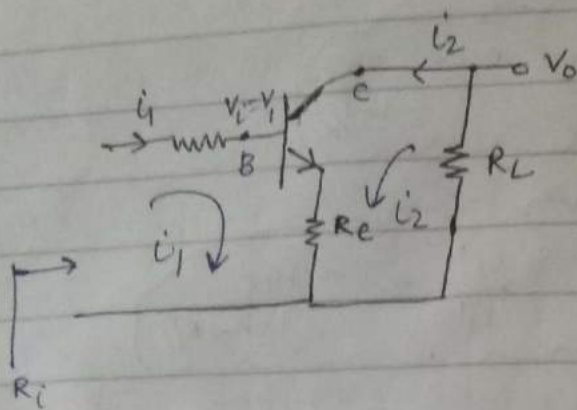


$$i_2 = \frac{R_2}{R_1+R_2} i_1$$



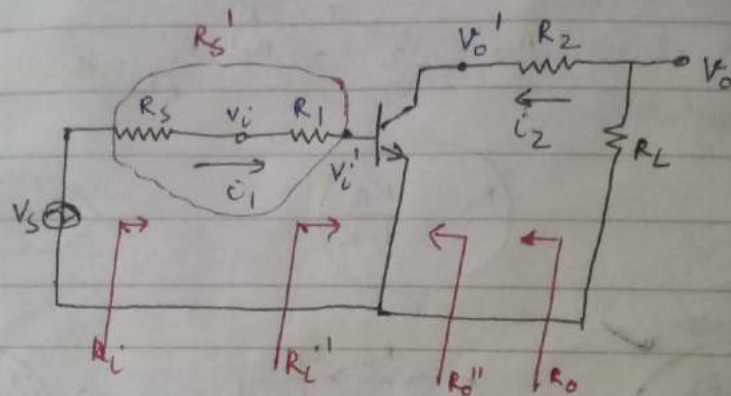
$$R_1 = R_E(1-A_I)$$

$$R_2 = R_E(1-\frac{1}{A_I})$$



$$K = \frac{i_2}{i_1} = A_I$$

$$A_V = \frac{V_o}{V_i} = R_L \cdot A_I = -\frac{i_2}{i_1}$$



$$A_I = -\frac{i_2}{i_1} = -\frac{h_{fe}}{1 + h_{oe} R_L}$$

$$R_i' = h_{ie} + A_I h_{oe} R_L'$$

$$R_i = R_i' + R_1$$

$$A_V' = \frac{V_o'}{V_i'} = A_I \cdot \frac{R_L'}{R_i'}$$

$$A_V = \frac{V_o}{V_i} = A_I \cdot \frac{R_L}{R_i}$$

$$Y_o'' = h_{oe} - \frac{h_{fe} \cdot h_{oe}}{R_S' + h_{ie}}$$

$$R_o'' = \frac{1}{Y_o''}$$

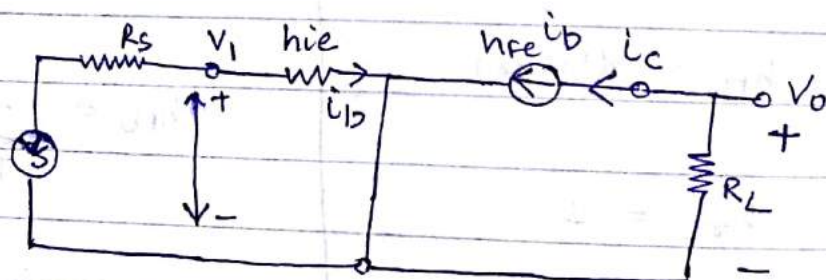
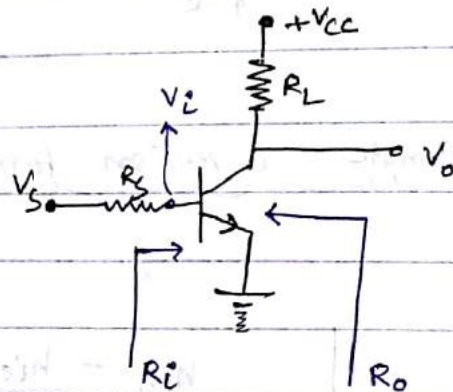
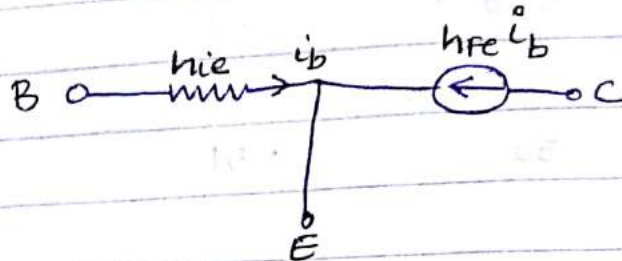
$$R_o'' + R_2 = R_o$$

$$R_L' = R_2 \parallel R_L$$

Resistance
 Gold coating $\rightarrow 5\%$
 Silver coating $\rightarrow 10\%$ (Tolerance)

Simplified CE model :-

$h_{oe} R_L \leq 0.1 \rightarrow \text{Condition}$



$A_I = -\frac{i_c}{i_b} = -h_{fe}$

error = $0.9.09$

$A_I = \left| \frac{-h_{fe}}{1 + h_{oe} R_L} \right|$

$R_i = h_{ie} + A_I h_{ie} R_L$

or

$R_i = h_{ie}$

error = 4.5%

$$A_{V(\text{app})} = \frac{A_{I(\text{app})} \times R_L}{R_{i(\text{app})}}$$

$$\frac{A'_{V(\text{app})}}{A_{V(\text{orig})}} = \frac{A_I R_L / R_i'}{A_I R_L / R_i}$$

$$= 1.1 \times \frac{R_i}{R_i'}$$

$$= 1.1 \times 1.045$$

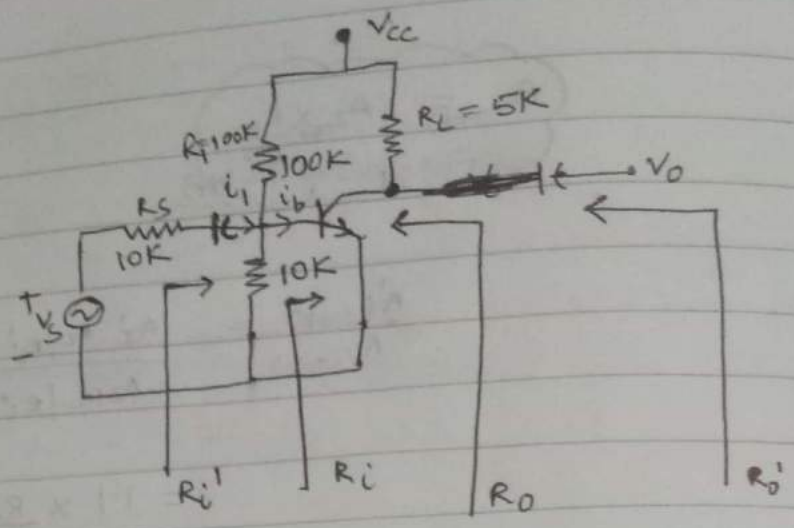
$$= \frac{1.495}{1.495} \quad \underline{A}$$

↓ error in A_V

74P(54.1)

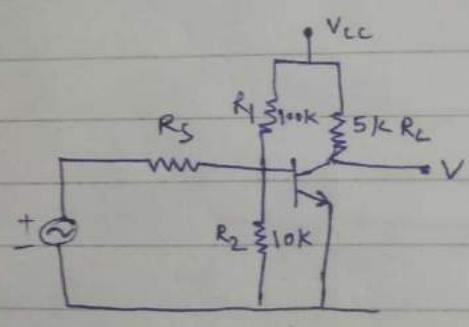
R_o

Que -

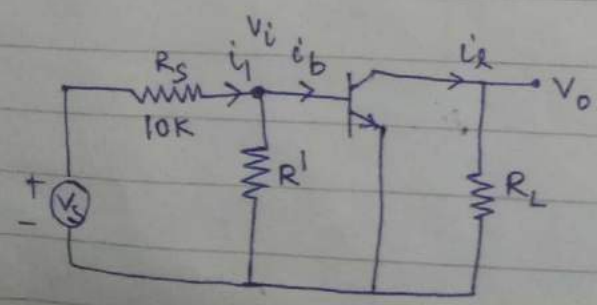


Find $A_I = \frac{i_2}{i_1}$, $A_V = \frac{V_o}{V_i}$, $A_{VS} = \frac{V_o}{V_s}$
 R_i, R_i', R_o, R_o'

Capacitor has removed for AC analysis.



Advantage
Disadvantage
Voltage divider
biasing



$$R_i' = R_1 || R_2 = 9.09K$$

$$A_I' = \frac{i_2}{i_b} = \frac{-h_{fe}}{1 + h_{oe} R_L}$$

$$A_I' = -44.44$$

$$R_o' = h_{ie} + h_{re} A_I' R_L$$

$$= 1.1K - 2.5 \times 10^{-4} \times 44.44 \times 5K$$

$$R_i = 1.04K\Omega$$

$$A_V = \frac{V_o}{V_i} = A_I' \times \frac{R_L}{R_i}$$

$$A_V = -213.653$$

$$R_o' = R_i \parallel R_i$$

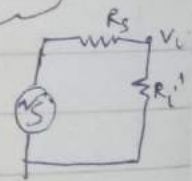
$$= 0.933K\Omega$$

$$A_{VS} = \frac{V_o}{V_S} = \left(\frac{V_o}{V_i} \right) \times \frac{V_i}{V_S}$$

$$A_{VS} = A_V \times \frac{R_o'}{R_o' + R_S}$$

$$A_{VS} = -18.232$$

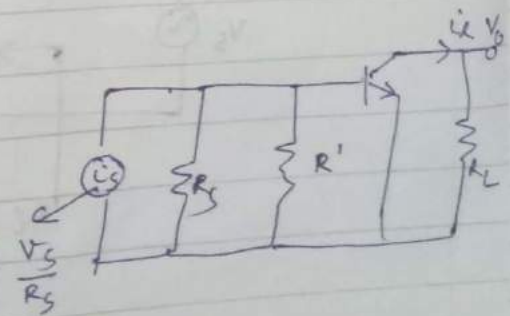
$$Y_o = h_{oe} - \frac{h_{fe} h_{re}}{R_S' + h_{ie}}$$



$$A_{IS} = \frac{I_o}{I_S}$$

$$A_{IS} = \frac{I_o}{I_S} \times \frac{I_o}{I_o} \times \frac{I_o}{I_o}$$

$$A_I = \frac{I_o}{I_i}$$



$\text{SiO}_2 \rightarrow$ Insulator
 bad conductor
 of electricity

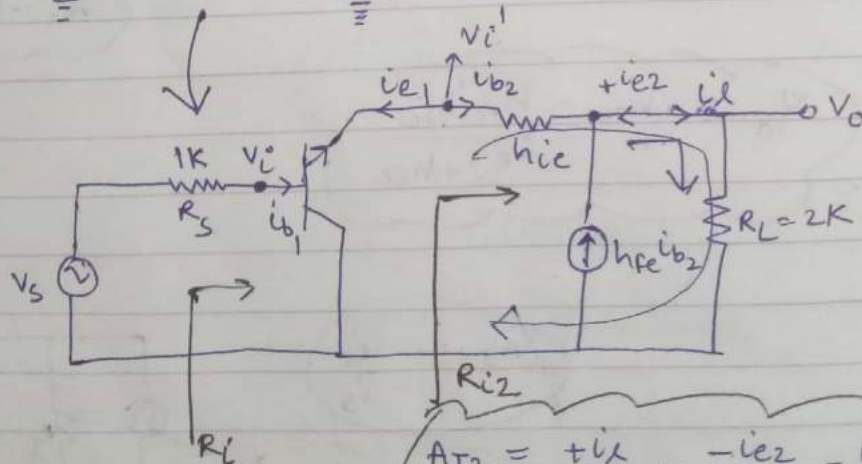
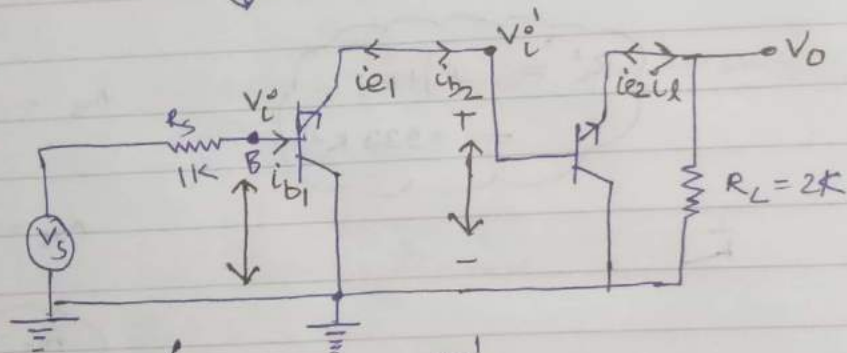
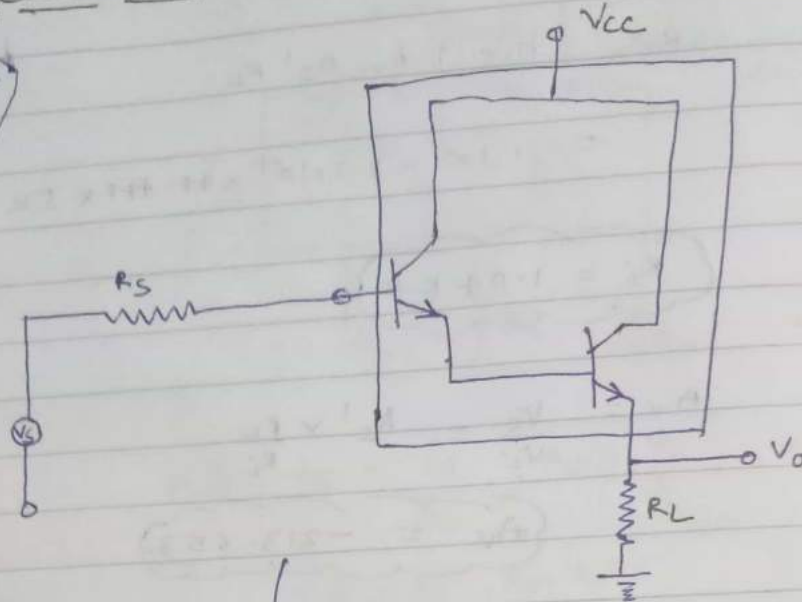
BJT $\rightarrow$ current controlled device
 FET $\rightarrow$ voltage controlled device

Onkar
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 Date: 12/02/18

High Input Impedance Circuit :-

Cascading
 Cascode

Common collector



$$A_{I2} = \frac{i_L}{i_{b2}} = \frac{-i_{e2}}{i_{b2}} = (1+h_{fe})$$

$$R_{i2} = \frac{V_{i'}}{i_{b2}} = h_{ie} + (1+h_{fe}) R_L$$

1034

"GRATITUDE IS THE BEST ATTITUDE"

$$A_{V2} = \frac{V_o}{V_i} = \frac{(1+h_{fe})\beta_2 R_L}{h_{ie}\beta_2 + (1+h_{fe})R_L\beta_2}$$

$$A_{V2} = \frac{(1+h_{fe})R_L}{h_{ie} + (1+h_{fe})R_L}$$

$$A_{V2} = 0.9893$$

For First Case;

$$\cancel{h_{oe}R_{i2} \leq 1}$$

$$A_{I1} = -\frac{i_{e1}}{i_{b1}} = \frac{-h_{fe}}{1+h_{oe}R_{i2}}$$

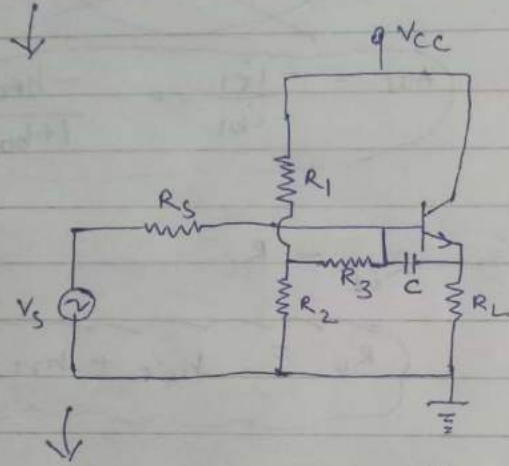
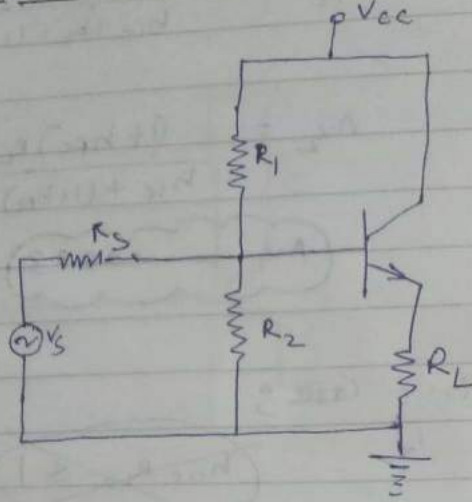
$$R_{i1} = R_i$$

$$R_i = h_{ie} + h_{oe} \cdot A_{I2} \cdot R_{i2}$$

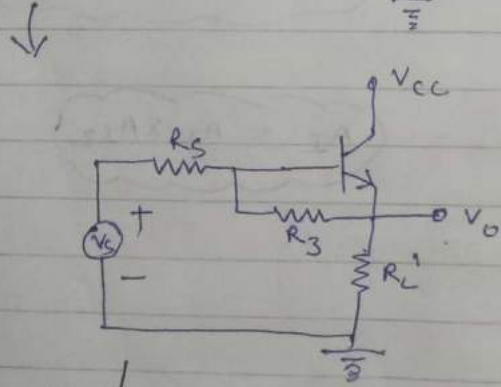
$$1952 \Omega \approx 1.5 M$$

$$A_I = A_{I1} \times A_{I2}$$

Boots strapping :-



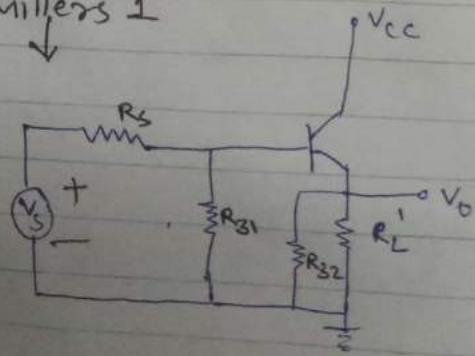
R_1, R_2, R_E etc
all are different
from previous one
but we took same name



$$R_L' = R_L || R'$$

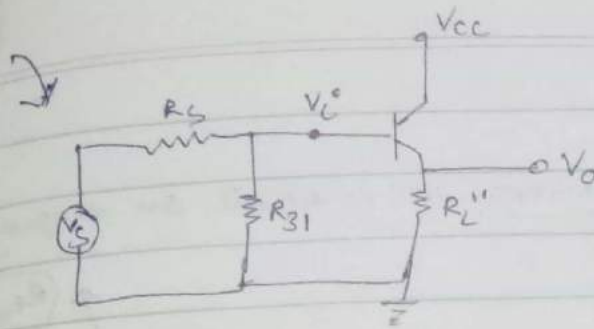
$$R' = (R_1 || R_2)$$

↓
Millers 1



$$R_{31} = \frac{R_3}{1 - A_V}$$

$$R_{32} = \frac{R_3}{1 - \frac{1}{A_V}}$$



$$R_L'' = R_{32} \parallel R_L'$$

$$A_v = \frac{V_o}{V_i} = K$$

$$R_{31} = \frac{2k}{1 - 0.995} = 4M$$

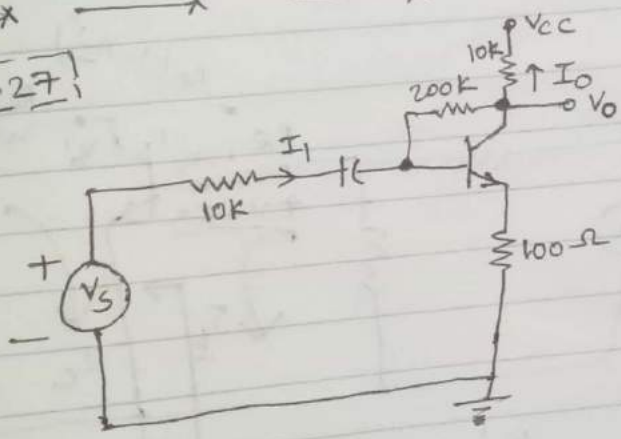
Suppose $A_v \approx 0.995$

$A_v \rightarrow \text{tends to } 1$
 $R_{31} \rightarrow \infty$

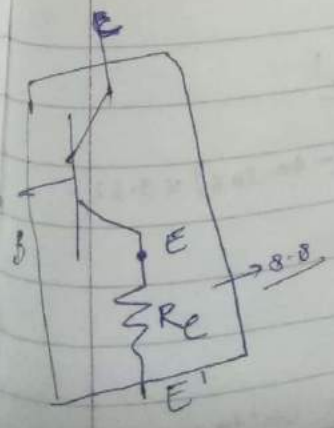
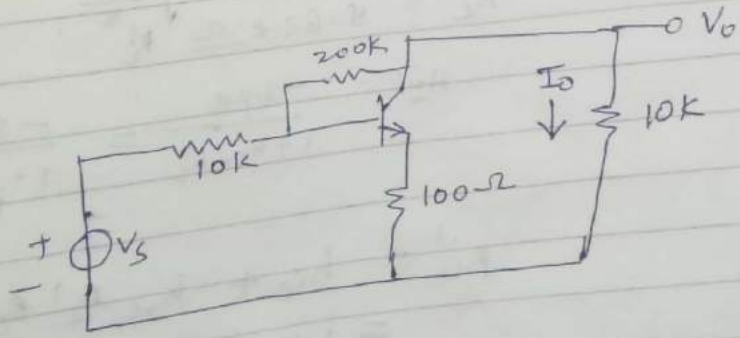
Now it will be right above for circuit.

Problem 8.27

Method 1



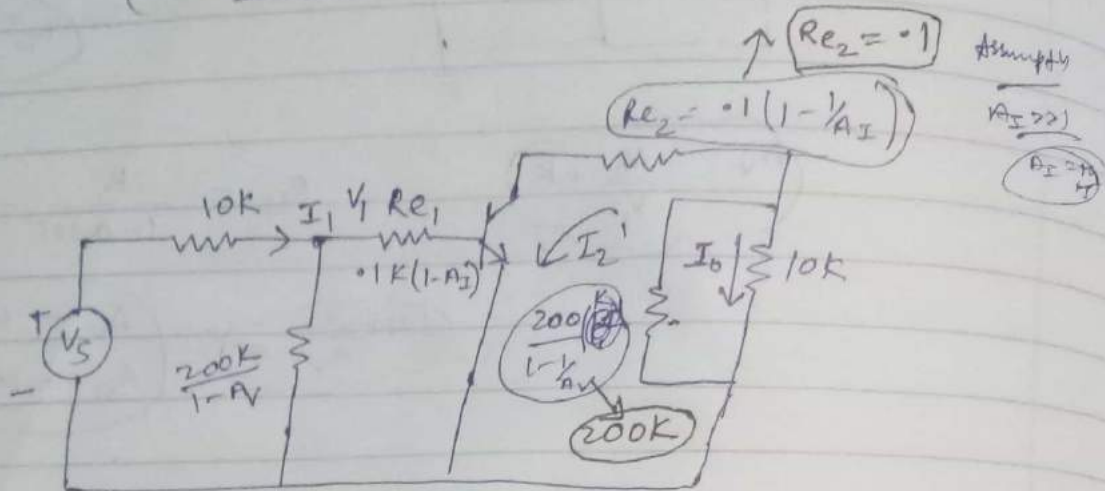
Common emitter configuration



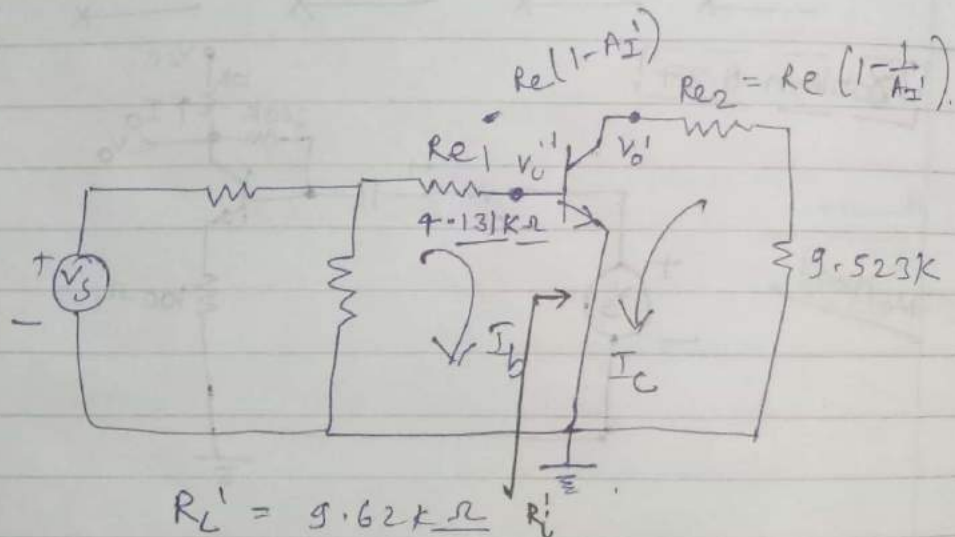
CE $(A_V \gg 1)$

Method 2

(Both millers will be activate on diagram)



Method 3



$$A_I' = \frac{-h_{fe}}{1 + h_{oe} R_L'} = \frac{-50}{1 + \frac{1}{50} \times 9.62} = -40.306$$

$$R_L' = h_{ie} + h_{oe} A_I' R_L'$$

$$= 1.1 + (2.5 \times 10^{-4}) \times (-40.306) \times 9.62K$$

$$= 1.003K\Omega$$

$$A_V' = A_I' \cdot \frac{R_L'}{R_i'}$$

new voltage divide